

Multi-Omics Analysis on AML Using MOFA

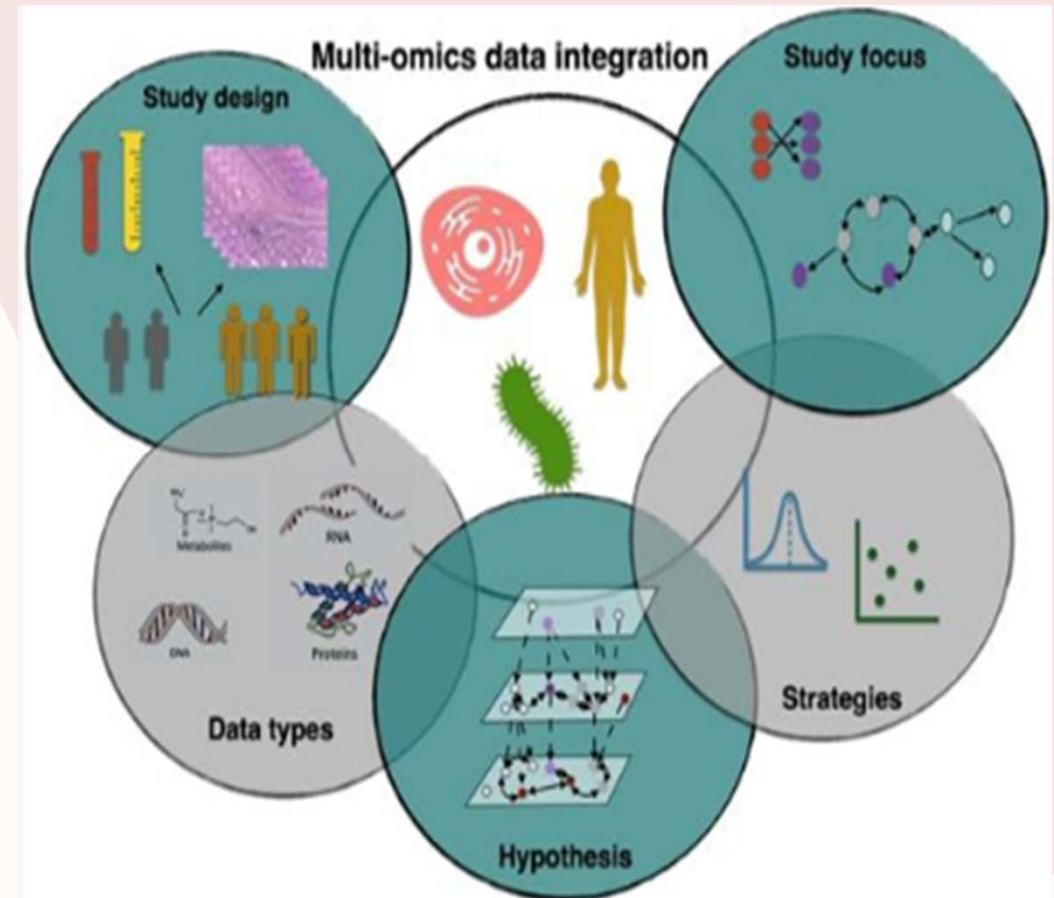
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INTRODUCTION

Multi-omics combines data from multiple platforms for comprehensive and in-depth profiling of biological processes. A critical parameter for maximizing the applicability of multi-omics profiling is data integration, which encompasses bioinformatic tools, pipelines, and databases. Data integration of multi-omics strategies promotes the development of biomarkers, definition of disease phenotypes, characterization of virulence factors, and discovery of druggable targets among other potential outcome.

Concept of Multi-omics data integration

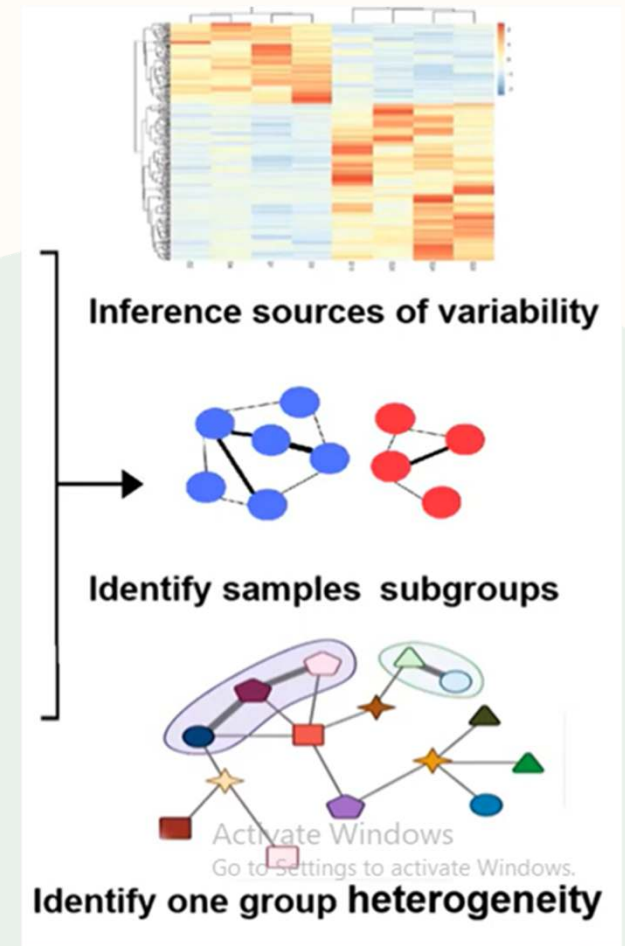
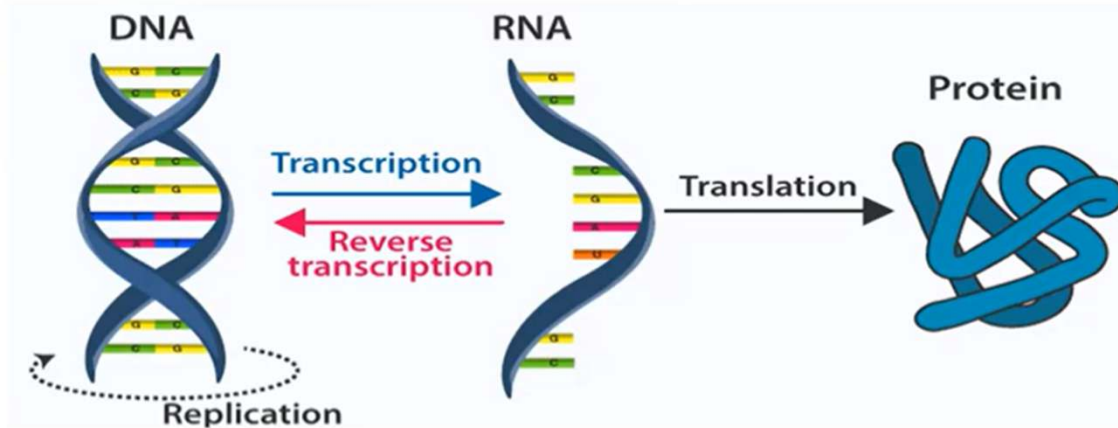


Purpose of multi-omics

1. Capturing the heterogeneity of data in one module or in one group or more.
2. Identifying missing data , unknown data and outliers.
3. Integrate different modules(different types of data).
4. Identifying sample subgroups.

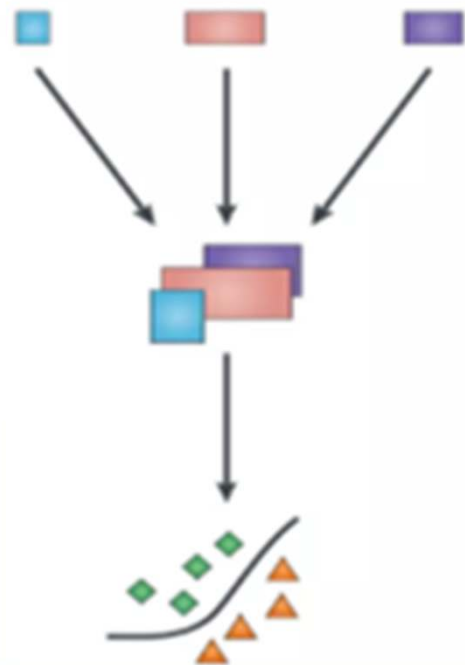
Purpose of multi-omics

Genomics Transcriptomics Proteomics

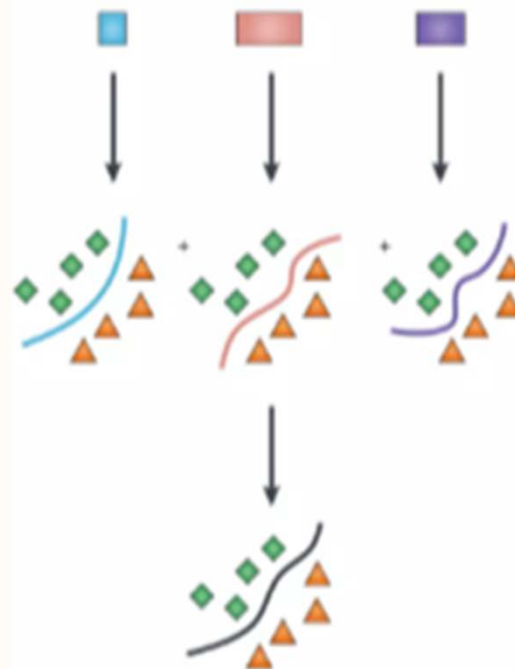


Types of integration methods

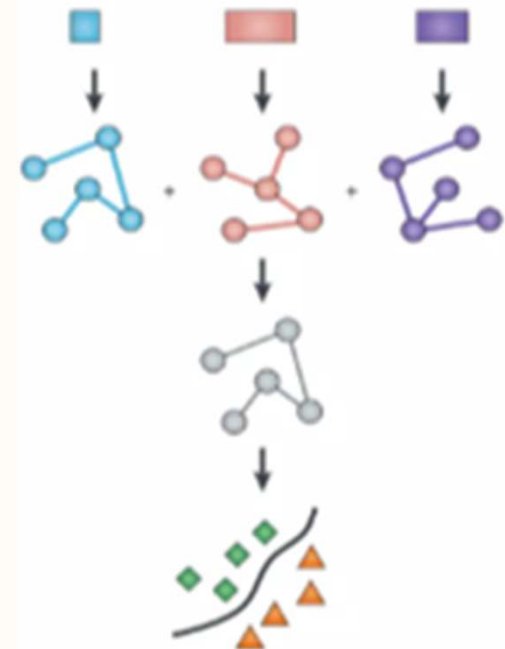
Concatenation-based integration



Model-based integration



Transformation-based integration



Multi-Omics Factor Analysis

Multi-Omics Factor Analysis

MOFA is a model that combines different types of biological data to find common patterns. Think of it like an advanced version of principal component analysis for multiple data sets. It identifies key factors that explain variations in various data types, helping to understand cellular states or disease groups.

Model overview of MOFA

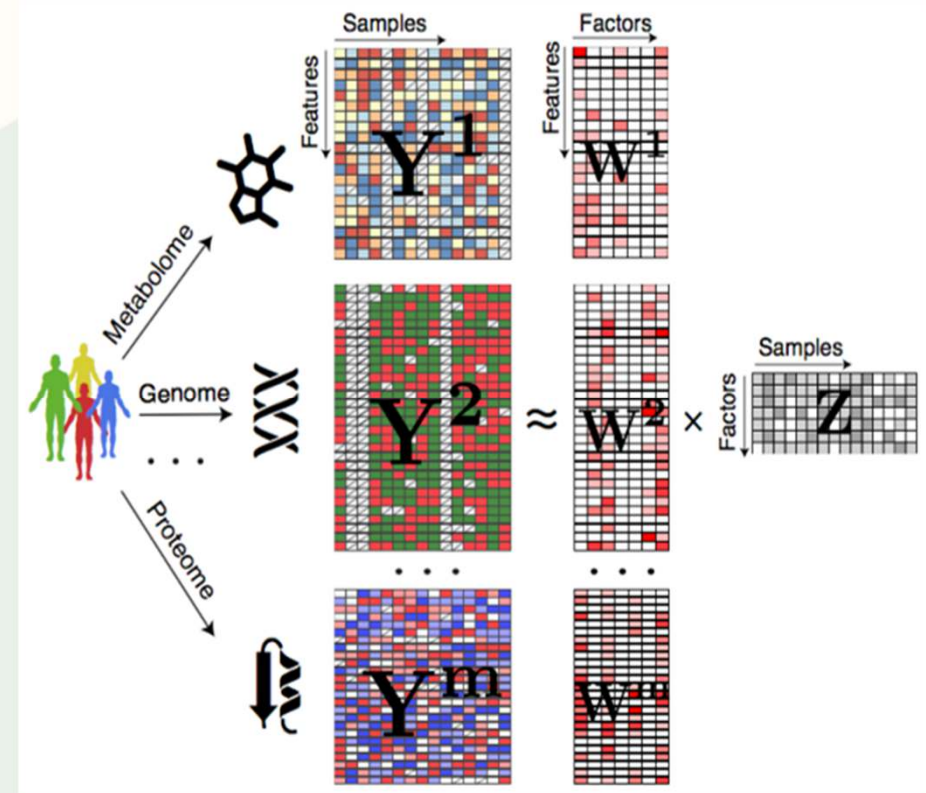


Fig 1: (a)

Why we used MOFA

Why we used MOFA

In general , We used MOFA to analyse a disease (Acute Myeloid Leukemia) to detect the status of each sample (sick or not sick).

- Contribution of MOFA :
MOFA contributes to benchmark our results to make sure of the downstream analysis of data .
- Comprehensive understanding of human complex diseases requires the analysis of multi-omics data reciprocally with clinical information.
- Based on the analysis, useful insights into the cellular functions are derived.
- The integration helps to understand the interplay of the biomolecules and in the assessment of the information from the omics data.

Further, it improves prognostic and predictive accuracies hence better treatment and prevention of the diseases

Why we used MOFA

MOFA (Multi-Omics Factor Analysis) is a tool specifically designed for integrating multiple types of omics data, such as genomics, transcriptomics, proteomics, and metabolomics. It has several advantages over other tools:

- Integration: MOFA combines diverse omics datasets into a single model, capturing shared and distinct variation across datasets
- Flexibility: MOFA can handle different data sizes and handle missing values, making it suitable for various multi-omics experiments.
- Dimensionality reduction: MOFA reduces the dimensionality of integrated data to decrease the attributes of the data , **can greatly enhance our understanding of biological systems when effectively integrated** and identifying key factors that explain the variation across datasets.

Why we used MOFA

- Visualization: MOFA provides interactive visualization tools to explore and interpret integrated data, helping identify patterns and relationships between datasets.
- Statistical modeling: MOFA incorporates statistical modeling for hypothesis testing and assessing the significance of patterns, enabling the discovery of biomarkers or predictive models.
- User-friendly: MOFA offers a user-friendly interface with tutorials, making it accessible to researchers with different computational expertise.

Acute myeloid leukemia

Acute Myeloid Leukemia (AML) is a cancer of the blood and bone marrow — the spongy tissue inside bones where blood cells are made. The word "acute" in Acute Myeloid Leukemia denotes the disease's rapid progression. It's called myeloid (myelos) leukemia because it affects a group of white blood cells called the myeloid cells.

we applied MOFA to a cohort of 200 patient samples of Acute Myeloid Leukemia (AML) that were profiled for miRNA, RNA-seq, and DNA methylation.

We have selected Acute Myeloid Leukemia due to our profound interest in leukemia, a field rich with extensive papers and numerous studies pertaining to this particular disease.

Applying MOFA on AML

We got the data of AML samples from TCGA (Linked omics).

We applied MOFA on this data by coding in R language and used DNA methylation, RNA-seq and micro RNA as views .

In AML ,MOFA discovers a low dimensional space spanned by known clinical markers and underappreciated axes of variation such as oxidative stress.

Researchers used MOFA to study Acute Myeloid Leukemia (AML) by combining clinical data like age and gender with genetic information. MOFA effectively handled different types of data, identifying shared and specific factors across them. It outperformed other methods in consistent factor identification, making it reliable for complex analyses. This integrated approach with diverse datasets ensured reliable comparisons and robust findings in the study of AML.

Omics types (data types)

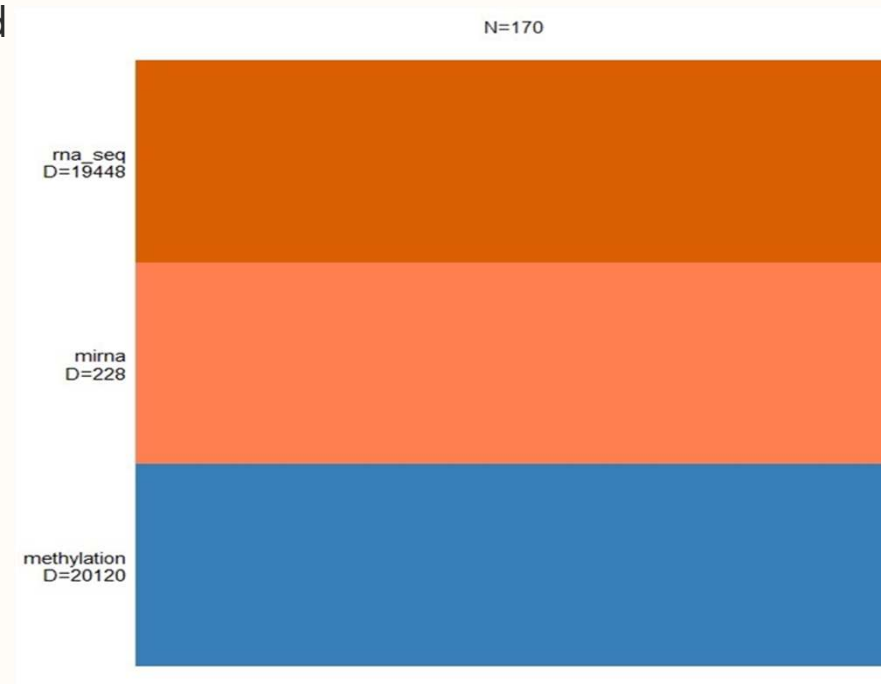


Fig 2

Knowing our variances

Here, we have broken down variances by omics type factor.

- We discovered that Factor 1 exhibits significant variation in DNA methylation modalities.
- Factor 2's RNA-Seq variance is moderate (medium).
- Factor 3 in RNA-seq and mirna are very similar. Because the variations of other components are expressed worse, we disregarded them.

Factors and variances

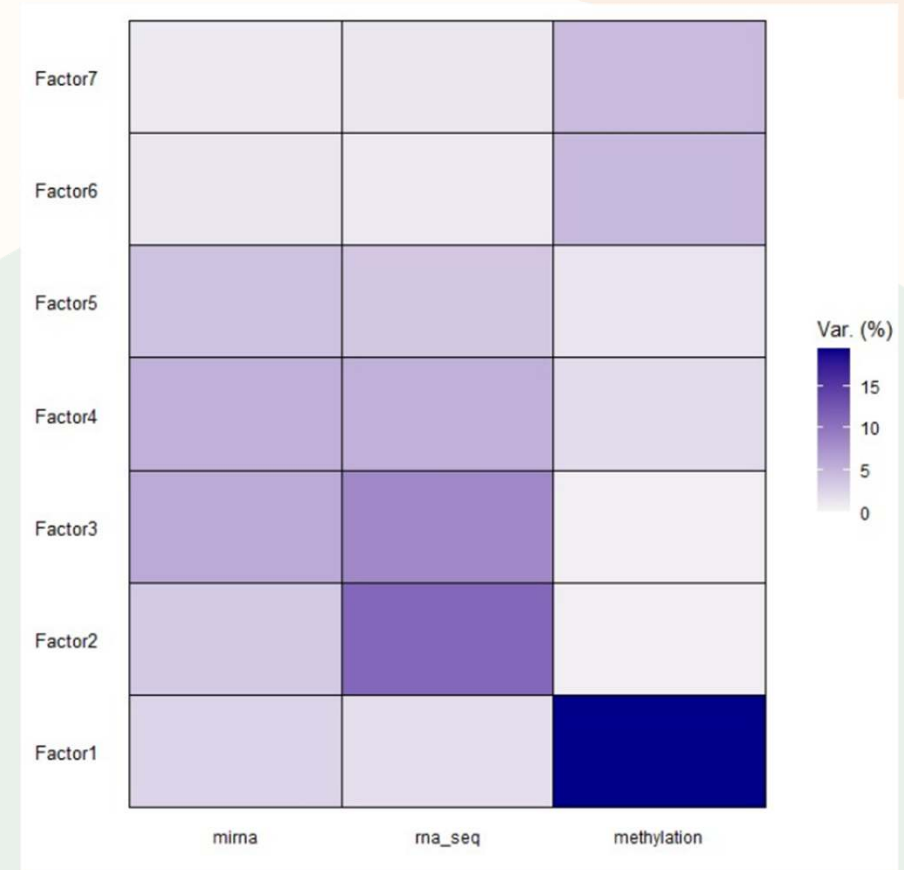
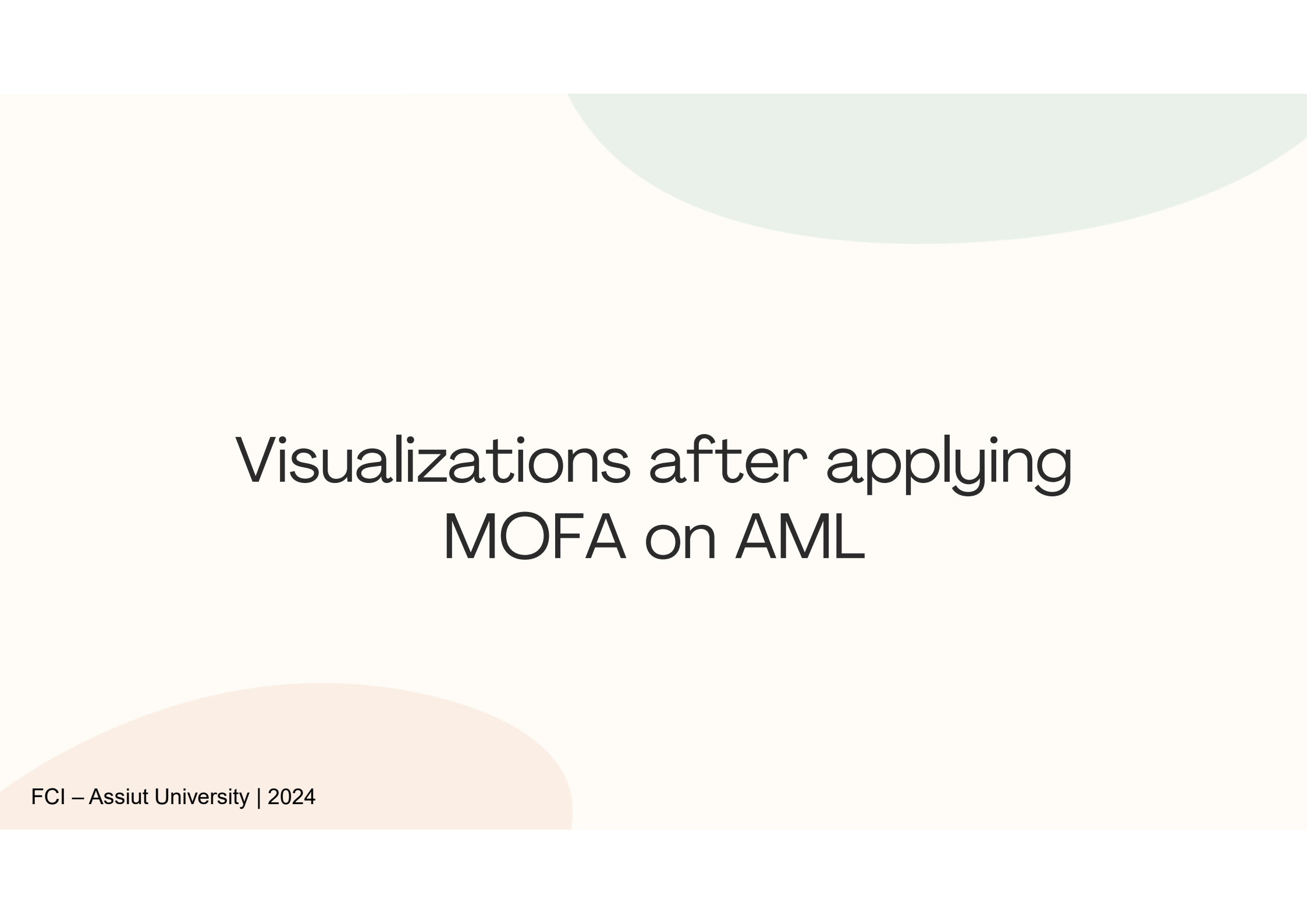


Fig 4



Visualizations after applying MOFA on AML

Visualization of factors

Figures 5a, 5b, and 5c effectively illustrate the distribution of clinical samples based on status, gender, and race, respectively, using color-coded dots to represent different groups.

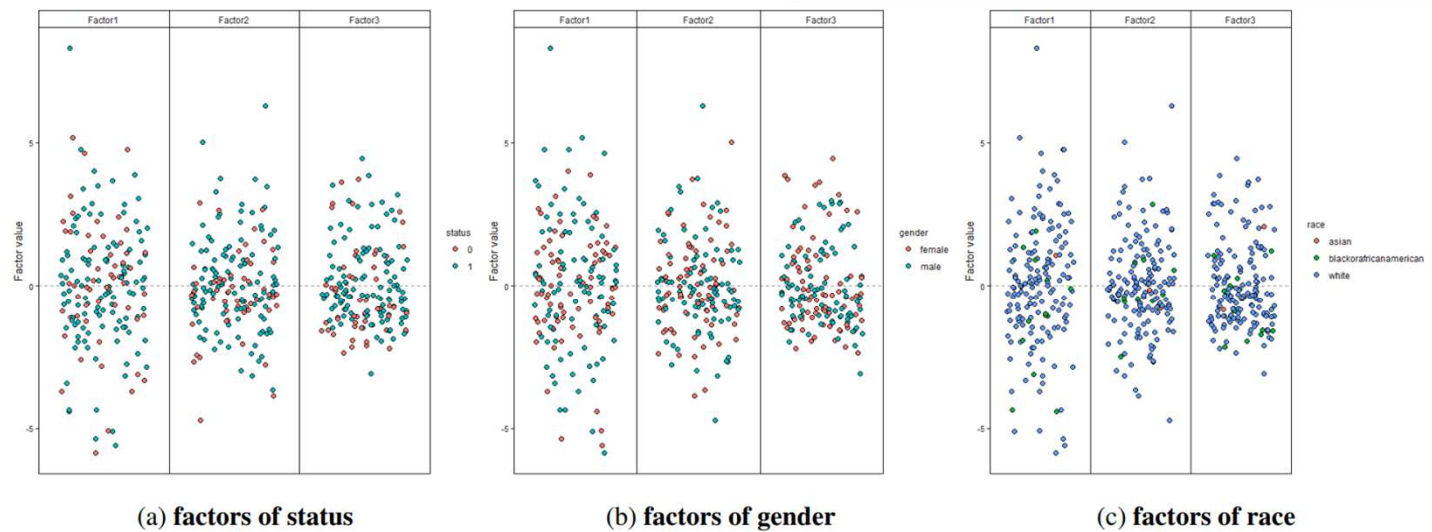


Fig. 5: The factors and their visualization

Visualization of factors cont.

These figures appear that the hidden layers in the data are represented by factors, and correlations between these factors and specific columns in the clinical data are being analyzed. Let's break down the paragraph to explain it further:

- In fig(a), the correlation analysis is focused on the "status" column in the clinical data. The factors, namely factor1, factor2, and factor3, are associated with this column. The correlation values range between -5 and 5, indicating the strength and direction of the relationship between the factors and the status column.

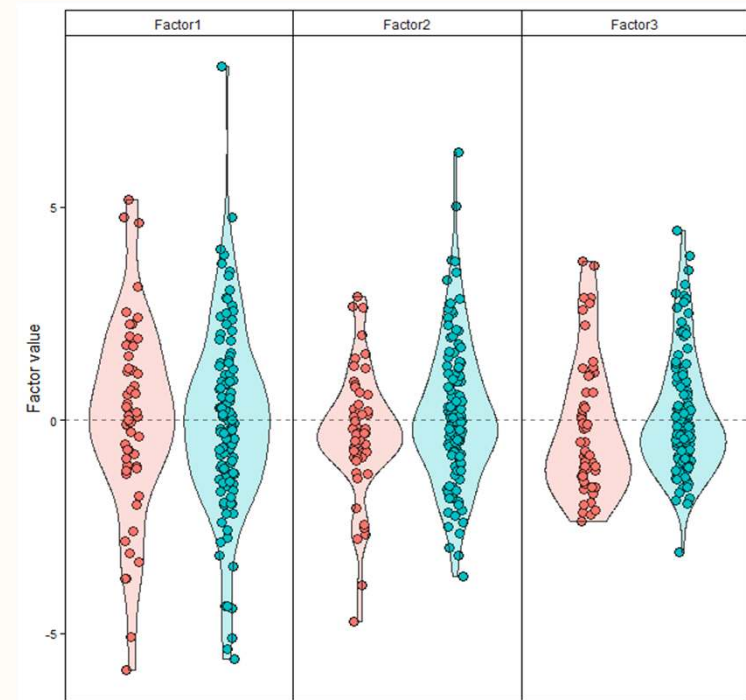


Fig 6: (a)

Visualization of factors cont.

- Moving on to fig(b), the correlation analysis is performed on the "gender" column. Three factors are considered in this case, although the specific factor names are not mentioned. The purpose is to explore the relationship between these factors and the gender column, providing insights into the hidden layers within the data.

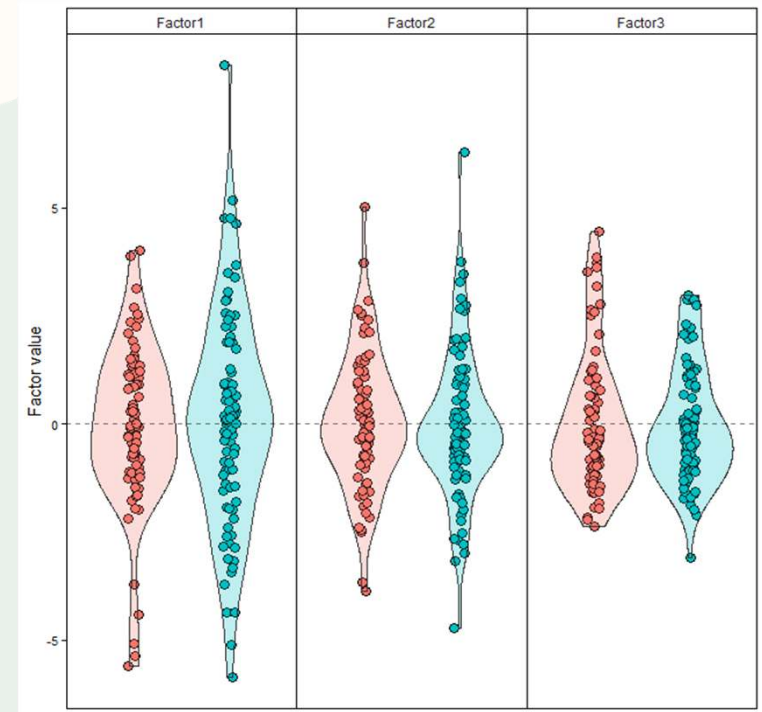


Fig 6: (b)

Visualization of factors cont.

- Similarly, fig(c) investigates the correlation between the "race" column and three factors. Again, the factor names are not specified, but the analysis aims to uncover the underlying structures or patterns within the data related to race.

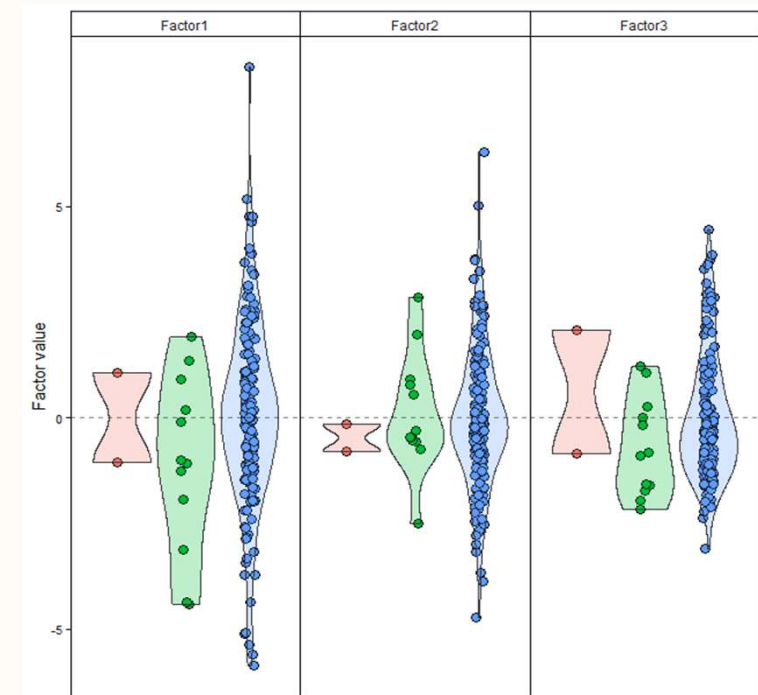


Fig 6: (c)

Visualization of features weights

the weights provide a score for how strong each feature relates factor 1. Methylation with no association with the factor 1

MOFA identifies important clinical markers in AMLas we have done factor 1 in datasets we found that the dna methylation has top weight in SBF1_view_3 genemirna has top weight in hsa.mir.181drna-seq has top weight in DNAJA4_view_2

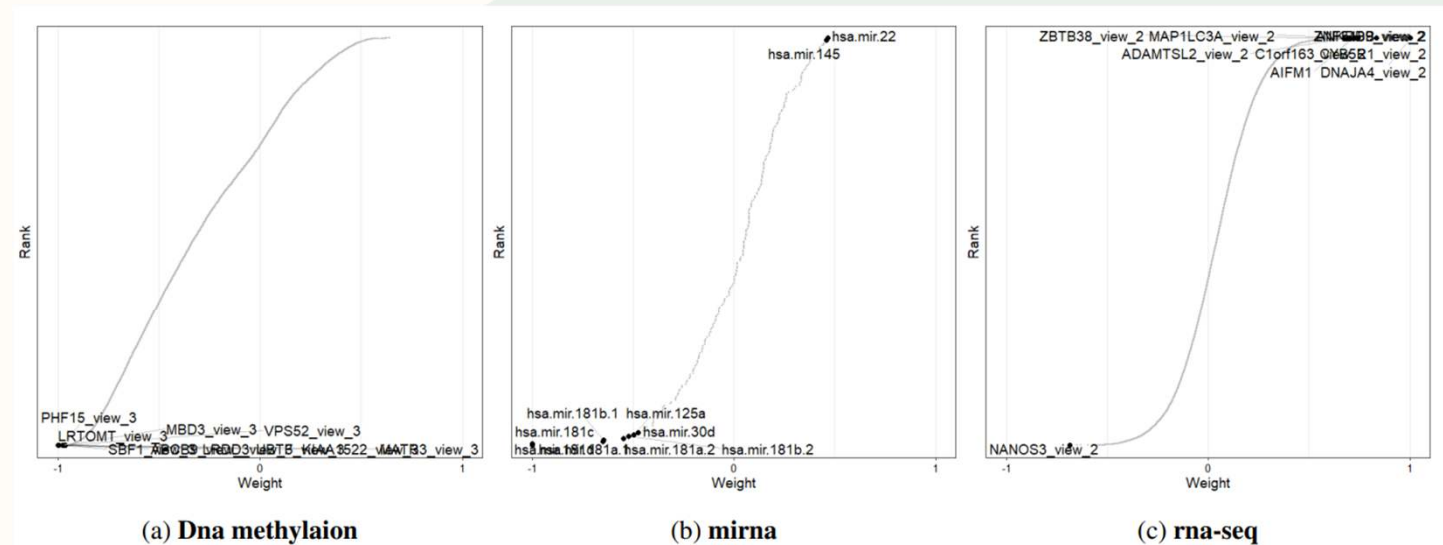


Fig. 7: The weights of features

Visualization of features weights cont.

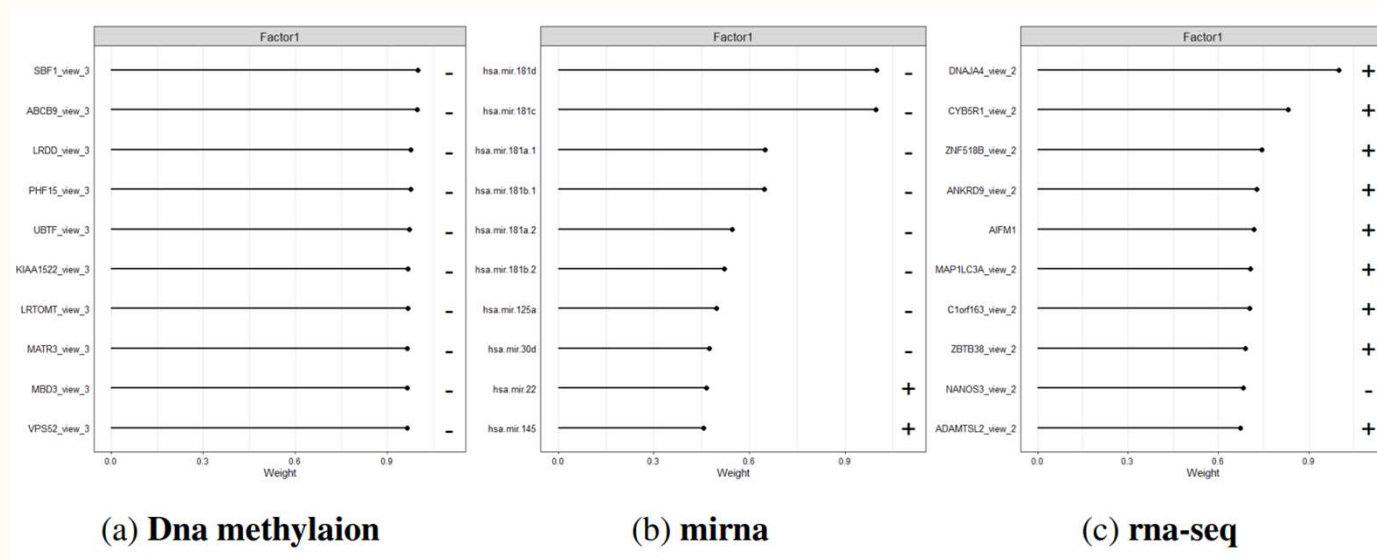


Fig 8: Top weights

Visualization of heatmap

Methylation Heatmap:

- Exploring methylation patterns across genders for selected genes.
- Heatmap depicting the methylation levels for specific genes and their association with gender.

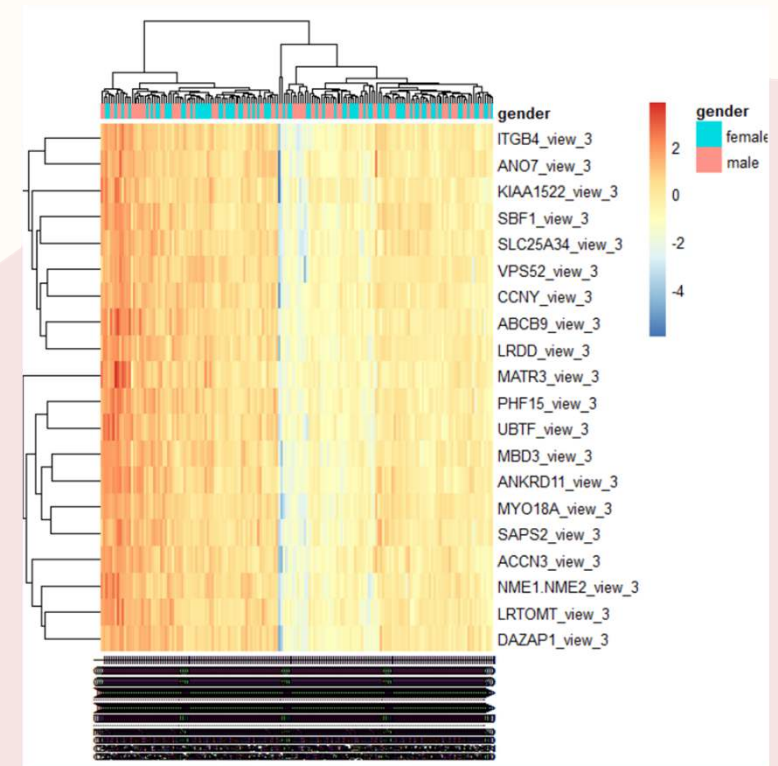


Fig 9: (a)

Visualization of heatmap cont.

miRNA Heatmap:

- Examining miRNA expression patterns based on status and gene names.
- Heatmap illustrating the miRNA profiles related to different statuses for the specified genes.

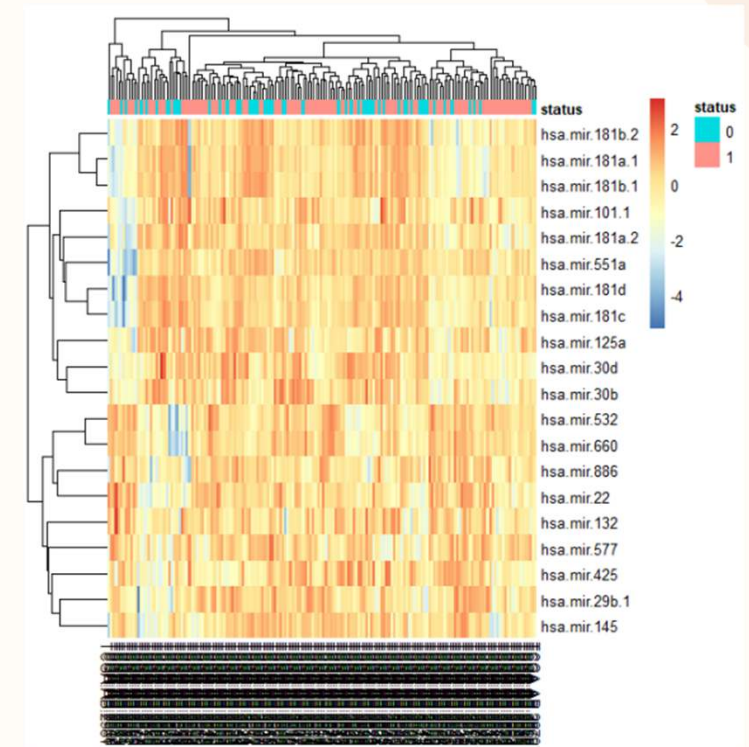


Fig 9: (b)

Visualization of heatmap cont.

RNA-seq Heatmap:

- Investigating RNA-seq data across ethnicities for the chosen set of genes.
- Heatmap showcasing the gene expression variations in RNA-seq data based on ethnicity.

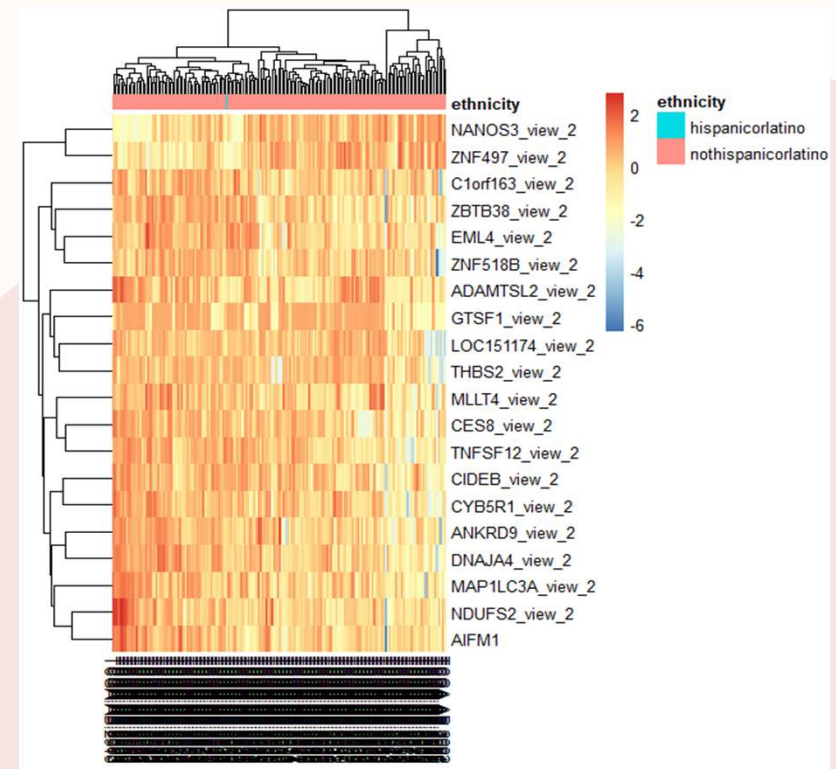


Fig 9: (c)

Visualization of Scatter plots

Methylation Scatter plot:

- Scatter plot representing the relationship between methylation levels, gender, and selected genes.
- Visualizing the scatter plot of methylation data, highlighting gender-specific patterns for certain genes.

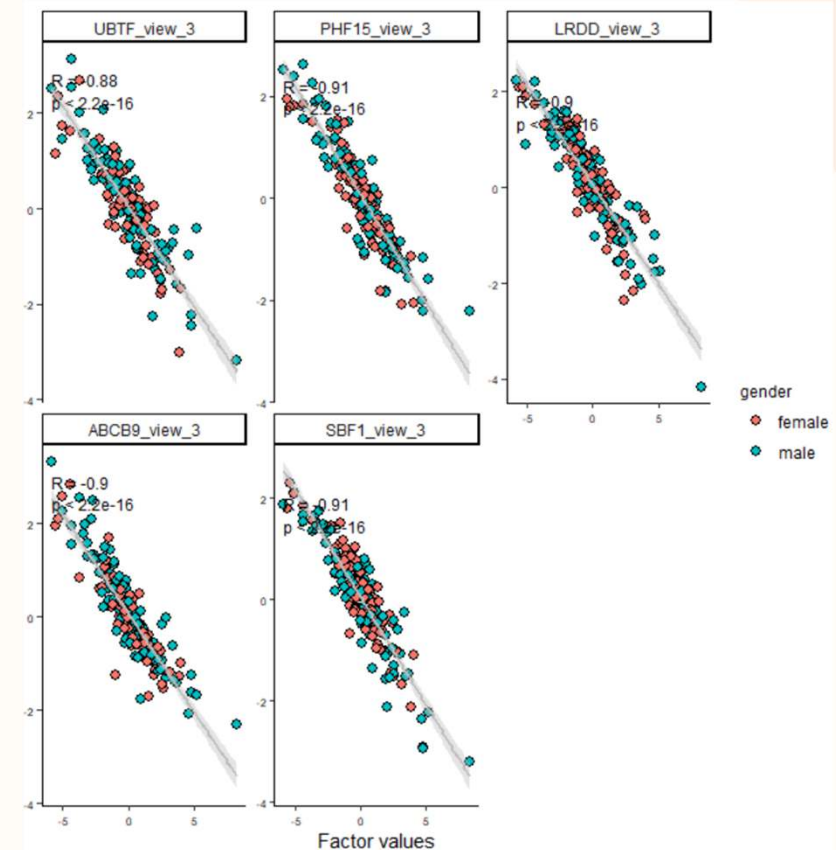


Fig 10: (a)

Visualization of Scatter plots cont.

miRNA Scatter plot:

- Scatter plot revealing the connection between miRNA expression, status, and gene names.
- Analyzing the scatter plot of miRNA data, emphasizing the correlation with different statuses for specific genes.

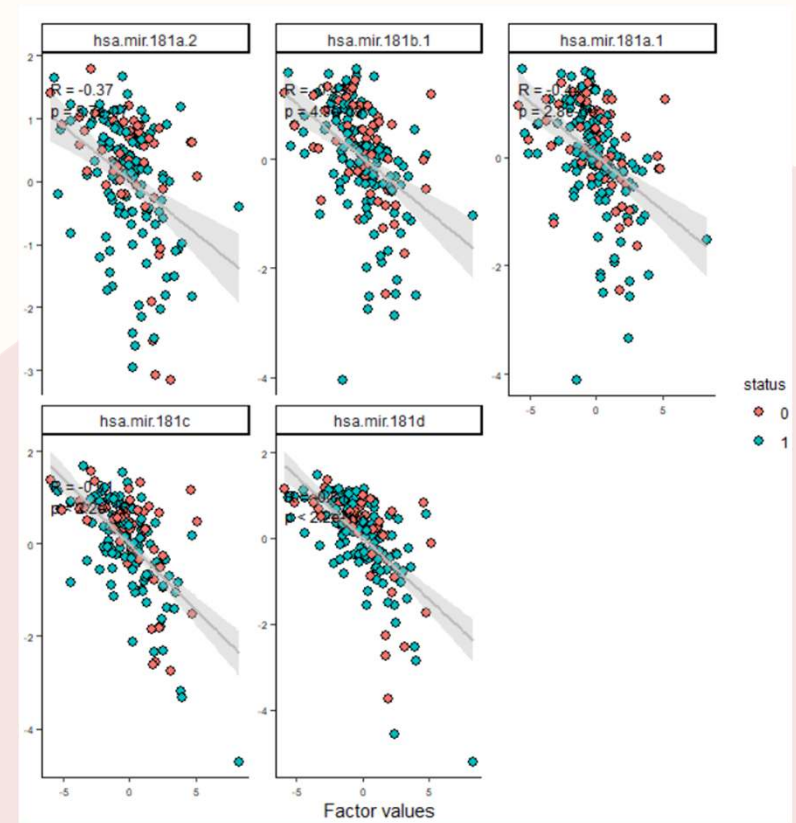


Fig 10: (b)

Visualization of Scatter plots cont.

RNA-seq Scatter plot:

- Scatter plot illustrating RNA-seq data, showcasing the interplay between ethnicity and gene expression.
- Exploring the scatter plot of RNA-seq data, emphasizing the impact of ethnicity on the expression of selected genes.

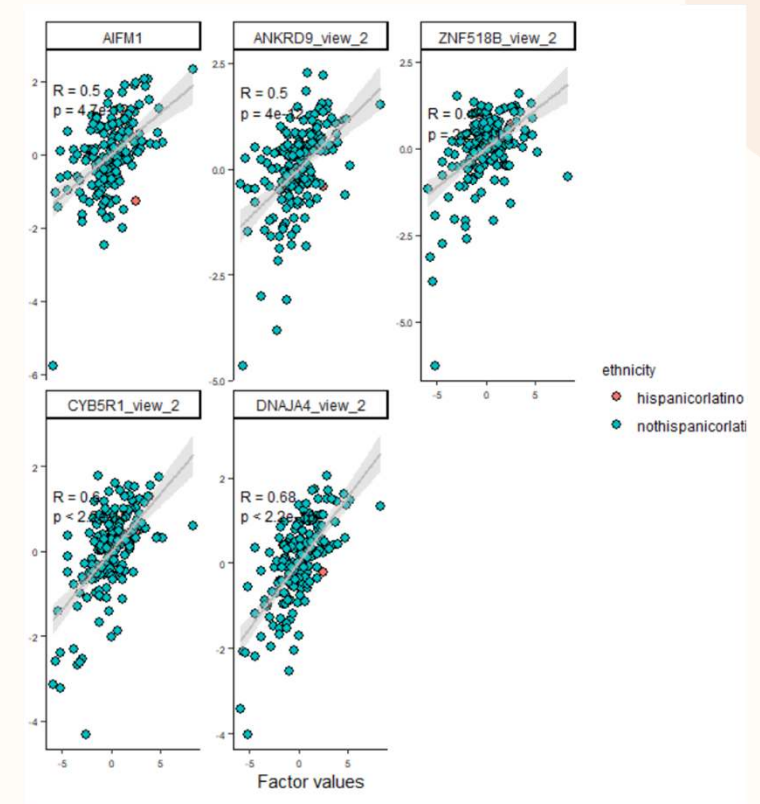


Fig 10: (c)

Conclusion

Multi-omics studies enrich our grasp of biological processes across diverse molecular layers. Nevertheless, the challenge lies in effectively integrating heterogeneous datasets from these studies. Addressing this gap, we present Multi-Omics Factor Analysis (MOFA), a computational tool unveiling primary sources of variation in multi-omics datasets.

MOFA deploys statistical modeling to infer hidden factors capturing both biological and technical variability. It discerns shared heterogeneity across modalities and specificity to individual data types. Learning these factors empowers downstream analyses, including subgroup identification, missing data imputation, and outlier detection.

In our AML patient study, MOFA was applied to a cohort of 200 samples, profiling miRNA, RNA-seq, and DNA methylation.

Link of our code :https://drive.google.com/drive/u/0/folders/1odUuYqc_rV7ACmGFUI4Ar3OeVEvDqyfx

Future work



Integrating Mutation Dataset

Since the mutation dataset includes a distinct marker that makes our data easier to understand, we will add it to the existing dataset.

Creating Network

For every omic data that we use, a network will be created.

Thank you!

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